

Complete Feature Checklist for Frontend Developer

1. User Authentication & Access Control

- Secure login page with username/password
- Role-based access (Admin, Manager/Supervisor, Operator/Staff)
- User account management interface
- Session management
- Activity logs viewer

2. Material Receipt Module

- Form to record incoming materials with:
 - Material type dropdown (PP Injection, HDPE Blow, LD, Rafia Pellets)
 - Quantity received (KG)
 - Quantity sent by supplier
 - Purchase price per KG
 - Total cost
 - Supplier/Sender details
 - Receipt date & time picker
- **Auto-calculations display:**
 - Transport loss quantity (KG)
 - Transport loss cost (₹)
 - Effective cost per KG after losses
- Photo upload functionality
- Batch ID generation (unique identifier)
- Approval workflow interface

3. Processing Workflow Module (4 Stages)

Stage 1: Sorting

- Input material quantity field
- Output quantity after sorting
- Auto-calculate loss (KG and %)
- Quality grading dropdown (Grade A, B, C, Rejected)
- Operator selection
- Notes/comments field
- Photo upload

Stage 2: Crushing

- Input from sorting output (auto-populated)
- Output quantity field
- Loss calculation display
- Processing notes field
- Photo upload

Stage 3: Washing

- Input from crushing (auto-populated)
- Output after washing field
- Loss tracking display
- Processing details field
- Photo upload

Stage 4: Pelletizing

- Input from washing (auto-populated)
- Final product quantity
- Loss at final stage
- Finished product specifications
- Photo upload

Loss Tracking Dashboard for all stages:

- Loss in KG and % for each stage
- Loss cost calculation
- Cumulative loss from receipt to finished product
- Loss history table

4. Sorting Module (Detailed)

- Interface to categorize mixed materials
- Quality grade assignment interface
- Rejected/defective materials tracking
- Process documentation with photos
- Operator and timing details
- Sorting status indicators

5. Inventory Management

Finished Goods Inventory Page:

- Product listing table with:
 - Product name/code
 - Current stock quantity
 - Unit type (KG, Pieces, Bags dropdown)
 - Unit price
 - Stock status badge
 - Stock value

Inventory Operations:

- Auto stock addition from processing
- Manual stock deduction form
- Stock adjustment form (with approval)
- Stock history log viewer

Stock Monitoring:

- Low stock alerts (visual indicators)
- Critical stock notifications
- Empty stock warnings
- Reorder reminders
- Product-wise inventory summary cards

6. Dashboard & Analytics

Main Dashboard Cards:

- Total material received (current month/YTD)
- Material processed by stage
- Finished products produced
- Current inventory value
- Monthly income
- Monthly expenses
- Balance summary

Key Visualizations (Charts):

1. **Loss Per Material Graph:**
 - Bar/line chart comparing losses at each stage
 - Grouped by material type
 - Sortable by highest loss
2. **Material Flow Chart:**
 - Sankey or flowchart showing material path
 - Quantity display at each stage
 - Loss highlights
3. **Inventory Status:**
 - Stock levels visualization
 - Low stock indicators
 - Inventory value pie chart
4. **Financial Summary:**
 - Daily/monthly income bar chart
 - Daily/monthly expenses chart
 - Profit/loss trend line
 - Cash position indicator

7. Income & Expense Management

Income Recording Page:

- Sales transaction form:
 - Customer details
 - Product sold dropdown
 - Quantity
 - Selling price
 - Total amount (auto-calculated)
 - Invoice number generator
 - Payment details

- Payment date picker
- Income transaction history table
- Filter by date range

Expense Recording Page:

- Expense form:
 - Expense type/category dropdown (Material costs, Transportation, Utilities, Labor, Equipment, Administrative, Other)
 - Amount field
 - Payment channel dropdown (Cash, GPay, Bank Transfer, Cheque)
 - Date picker
 - Description textarea
 - Document upload (receipts/bills)
- Expense transaction history table
- Category-wise expense summary

8. Financial Reports

Balance Sheet Page:

- Assets section:
 - Cash
 - Bank balance
 - Inventory value
- Liabilities section
- Equity overview
- Financial position summary
- Date range filter

Profit & Loss Statement:

- Total revenue
- Total expenses breakdown
- Gross profit
- Net profit/loss
- Profit margins display
- Period comparison

Cash Flow Report:

- Cash inflows table
- Cash outflows table
- Net cash flow calculation
- Ending cash position
- Visual cash flow chart

Bank Account Management:

- Multiple bank accounts table

- Balance by account
- Transaction history per account
- Account summary cards
- Add/edit bank account form

9. Loss Analysis & Reporting

Loss Tracking Page:

- Total loss by stage (table view)
- Loss by material type (grouped view)
- Cost impact per loss
- Cumulative loss display
- Loss trends chart (over time)

Loss Reports:

- Monthly loss summary
- Loss percentage by stage
- Trend analysis (improvement indicators)
- Comparison with previous periods
- Downloadable reports

10. Batch Tracking

Batch Tracking Interface:

- Batch ID search/filter
- Complete batch history timeline
- Stage-by-stage tracking view
- Source to finished product traceability
- Batch status indicators
- Detailed batch information card

11. Quality Management

Quality Dashboard:

- Grade assignment interface (A, B, C, Rejected)
- Quality metrics cards
- Defect documentation form
- Quality trends charts
- Material-wise quality summary

12. Document Management

Document Center:

- Centralized document repository

- Upload interface with categorization
- Document types: Travel documents, Receipts, Process docs, Transaction proofs
- Search and filter functionality
- Document preview
- Download options
- Folder structure by category

13. Data Export & Reporting

Export Center:

- Export buttons for:
 - Balance sheet (Excel/CSV/PDF)
 - Income statement (Excel/CSV/PDF)
 - Expense reports (Excel/CSV/PDF)
 - Inventory reports (Excel/CSV/PDF)
 - Loss analysis reports (Excel/CSV/PDF)
- Custom date range selector
- Export format dropdown
- Scheduled exports setup
- Email report functionality

14. Alerts & Notifications

Notification System:

- Notification bell icon with counter
- Notification panel with:
 - Low stock alerts
 - Critical inventory warnings
 - Approval requests
 - Daily summary
- Mark as read functionality
- Notification preferences page
- Email notification settings

15. Audit Trail

Audit Log Page:

- Activity log table:
 - User name
 - Action performed
 - Timestamp
 - Details/description
 - Change history
- Filter by user, action type, date
- Search functionality
- Export audit logs

16. User Management (Admin Only)

User Management Page:

- User list table
- Add new user form
- Edit user details
- Role assignment
- Activate/deactivate users
- Reset password functionality
- User activity history

17. System Configuration (Admin Only)

Settings Page:

- Material types configuration
 - Processing stage settings
 - Alert threshold settings
 - Email notification setup
 - System preferences
 - Backup settings
 - Company information
-

UI/UX Requirements:

1. **Responsive Design:** Mobile-friendly interface
2. **Navigation:** Sidebar with role-based menu items
3. **Color Scheme:** Professional, clean interface
4. **Loading States:** Spinners/skeletons for data loading
5. **Error Handling:** User-friendly error messages
6. **Success Messages:** Toast notifications for actions
7. **Confirmation Dialogs:** For critical actions (delete, approve)
8. **Date Pickers:** Consistent date selection components
9. **File Upload:** Drag-and-drop with preview
10. **Search/Filter:** On all list/table views
11. **Pagination:** For large datasets
12. **Breadcrumbs:** For navigation clarity